EXP 1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

AIM

To implement and demonstrate the FIND-S Algorithm for finding the most specific hypothesis based on a given set of training data samples.

SOFTWARE USED

- Google Colab
- Python 3.x
- Pandas Library

PROCEDURE

1. Import the Pandas library.
2. Create the training dataset.
3. Initialise the hypothesis using the first positive example.
4. Compare the hypothesis with all positive examples.
5. Replace differing attribute values with.
6. Ignore negative examples.
7. Display the final hypothesis.

CODE

```
import pandas as pd
# Create the training dataset
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

columns = ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'EnjoySport']
df = pd.DataFrame(data, columns=columns)

print("Training Dataset:\n")
print(df)

# Find-S Algorithm
hypothesis = None

for index, row in df.iterrows():
    if row['EnjoySport'] == 'Yes':
        if hypothesis is None:
            hypothesis = list(row[:-1])
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != row[i]:
                    hypothesis[i] = '?'
print("\nMost Specific Hypothesis:")
print(hypothesis)
```

OUTPUT

```
Training Dataset:

Sky   Air    Temp    Humidity    Wind     Water     Forecast   Enjoy Sport
0     Sunny    Warm   Normal    Strong    Warm     Same       Yes
1     Sunny    Warm    High    Strong   Warm     Same       Yes
2     Rainy    Cold     High      Strong    Warm   Change       No
3     Sunny    Warm    High    Strong    Cool   Change      Yes

Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
/tmp/ipykernel_1084/4257998243.py:27: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behaviour). To access a value by position, use `ser.iloc[pos]`
  if hypothesis[i] != row[i]:
```

RESULT

The FIND-S algorithm was successfully implemented in Google Colab, and the most specific hypothesis obtained is:

['Sunny', 'Warm', '?', 'Strong', '?', '?']